

DS Practical - 6

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PART – B : Question

Tree Traversal: Write a program to:

a. Implement pre-order

CODE :

```
class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def preorder(root):
    if root:
        preorder(root.left)
        print(str(root.val) + "->", end='')
        preorder(root.right)

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("preorder Traversal")
preorder(root)
print("\n KOMAL S049")
```

OUTPUT :

```
preorder Traversal
4->2->5->1->3->
KOMAL S049
```

b. In-order

CODE :

```
class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def inorder(root):
    if root:
        inorder(root.left)
        print(str(root.val) + "->", end='')
        inorder(root.right)

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("Inorder Traversal")
inorder(root)
print("\n KOMAL S049")
```

OUTPUT :

```
-----
Inorder Traversal
4->2->5->1->3->
  KOMAL S049
```

c. Post-order traversal of a binary tree.

CODE :

```

class Node:
    def __init__(self, item):
        self.left = None
        self.right = None
        self.val = item

def postorder(root):
    if root:
        postorder(root.left)
        print(str(root.val) + "->", end='')
        postorder(root.right)

root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)

print("postorder Traversal")
postorder(root)
print("\n KOMAL S049")

```

OUTPUT :

```

postorder Traversal
4->2->5->1->3->
KOMAL S049

```